



使用 Kubean 管理集群生命周期

 DaoCloud 江波

bo.jiang@daocloud.io





Content 目录

01  背景介绍

02  功能原理

03  社区发展



Part 01



背景介绍



背景介绍

管理集群生命周期 所面临的问题

⚠ 难以支持国产
信创及异构环境

⚠ 复杂的运维流
程，操作易出错

⚠ 集群组件较
多，参数管理混
乱

⚠ 弱版本管理，
集群升级难度高

⚠ 操作无记录，
难以排障溯源

⚠ 无法应对离线
化的部署场景

背景介绍

寻找集群生命周期管理 最佳社区方案



Why kubespray ?

- 支持 10 余种主流 linux 发行版的部署
- 支持丰富的网络、存储、运行时组件扩展
- 支持扩容异构节点
- 支持完整的生命周期管理，且操作灵活
- 基于 ansible, 对于运维交付友好
- 良好的开源社区活跃度

What else is missing?

1. 基于声明式 API 的云原生操作模式
2. 集群管理操作编排的灵活扩展
3. 国产信创系统的广泛支持
4. 离线化的完备部署能力



Part 02

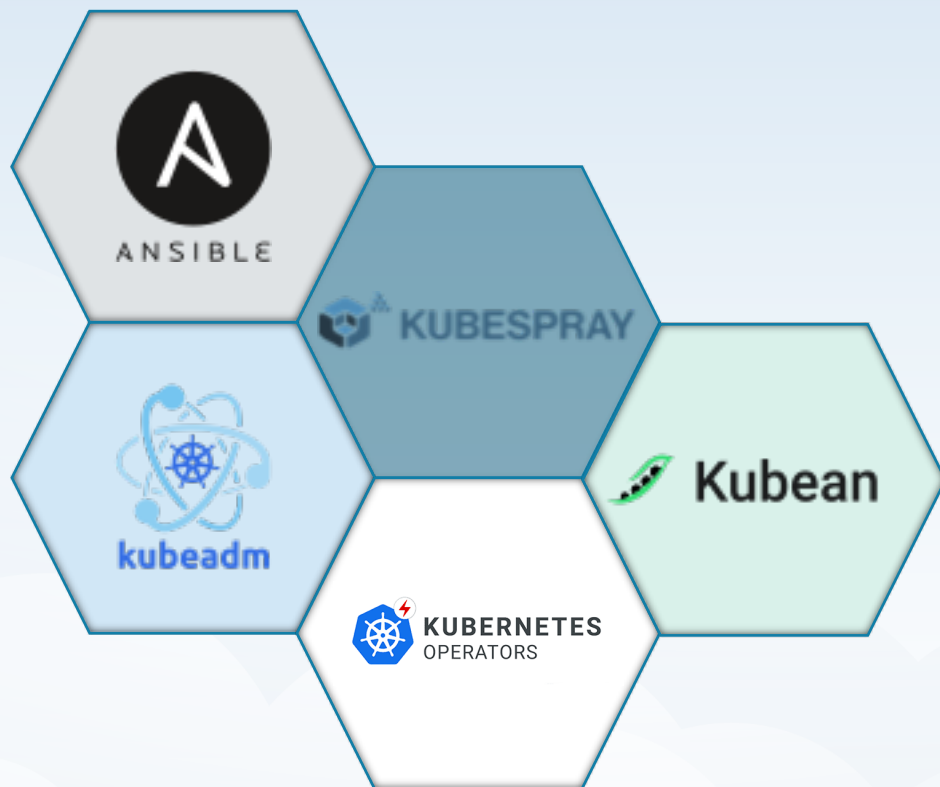


功能原理

功能原理

What is kubernetes ?

Kubean 是一款基于 kubespray 的集群生命周期管理工具，其目的是采用云原生的方式简化集群的部署及生命周期管理。

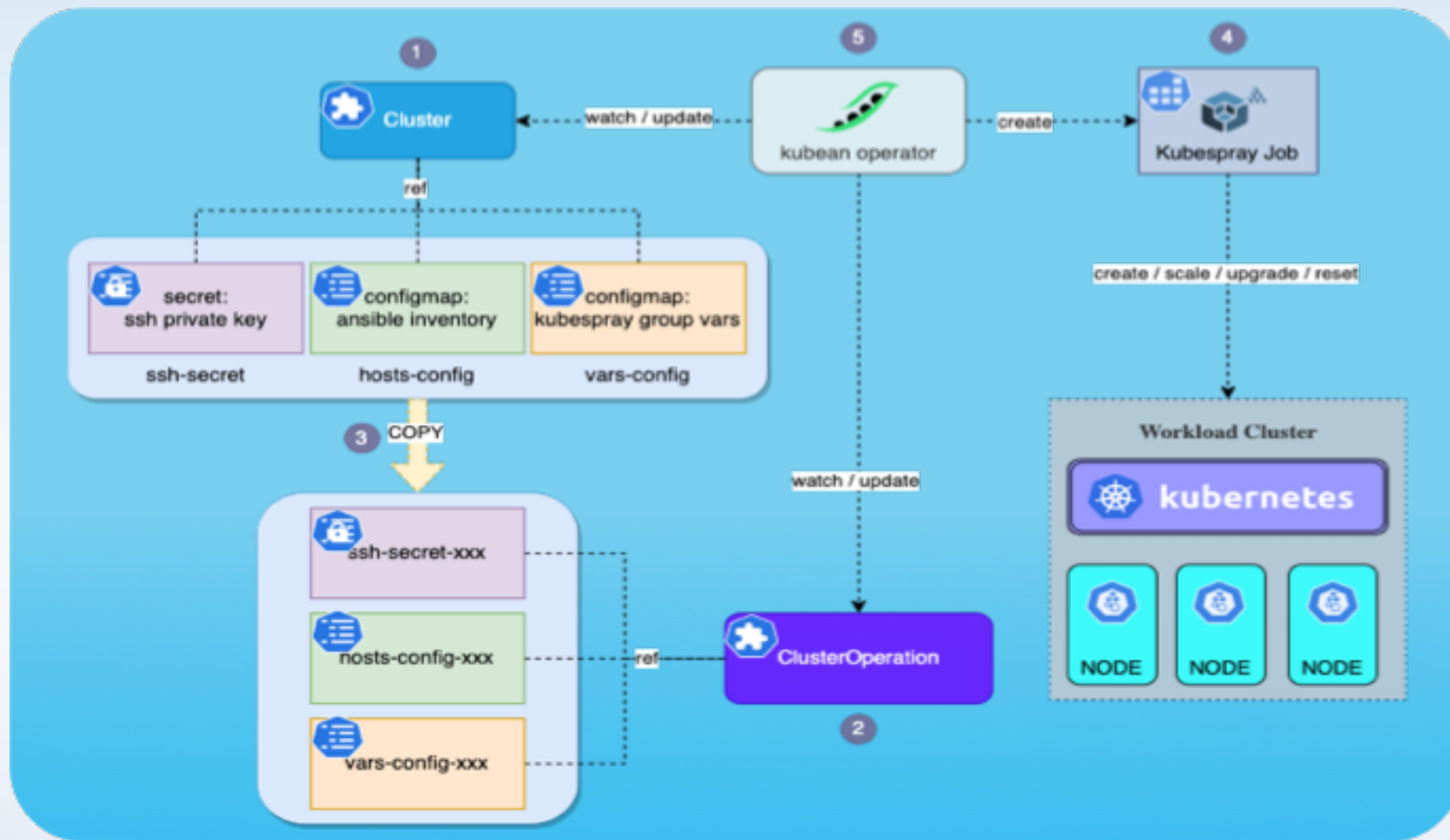


Awesome features

- 提供声明式 API 的云原生操作模式
- 提供离线场景的完备部署能力
- 允许操作任务编排的灵活扩展
- 支持国产信创系统及多架构场景
- 允许留存集群管理操作记录

🏗️ 功能原理

Kubean 基本架构原理



功能原理

🔧 Kubean 为集群操作编排提供了灵活的扩展能力：

- PreHook & PostHook 可在任务前后扩展操作内容
- Action Resource Ref 可自定义复杂任务的操作内容

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: cluster1-demo-myaction
  namespace: kubean-system
data:
  myplaybook.yml: |
    - hosts: k8s_cluster
      gather_facts: false
      become: yes
      any_errors_fatal: "{{ any_errors_fatal | default(true) }}"
      tasks:
        - name: Print inventory hostname
          debug:
            msg: "inventory hostname is {{ inventory_hostname }}"
  hello.sh: |
    echo "hello world!"

apiVersion: kubean.io/v1alpha1
kind: ClusterOperation
metadata:
  name: cluster1-demo-ops-1
spec:
  cluster: cluster1-demo
  image: ghcr.io/kubean-io/spray-job:latest
  actionType: playbook
  action: myplaybook.yml
  actionSource: configmap
  actionSourceRef:
    name: cluster1-demo-myaction
    namespace: kubean-system
  preHook:
    - actionType: shell
      action: hello.sh
      actionSource: configmap
      actionSourceRef:
        name: cluster1-demo-myaction
        namespace: kubean-system
```

```
apiVersion: kubean.io/v1alpha1
kind: ClusterOperation
metadata:
  name: cluster-ops-01
spec:
  cluster: sample
  image: ghcr.io/kubean-io/spray-job:latest
  actionType: playbook
  action: cluster.yml
  preHook:
    - actionType: playbook
      action: ping.yml
    - actionType: playbook
      action: enable-repo.yml
      extraArgs: |
        -e "{{ repo_list: ['http://10.20.30.40:9000/kubean/centos/${releasever}/os/${basearch}'] }}"
    - actionType: playbook
      action: disable-firewall.yml
  postHook:
    - actionType: playbook
      action: cluster-info.yml
    - actionType: playbook
      action: enable-repo.yml
      extraArgs: |
        -e undo=true
```

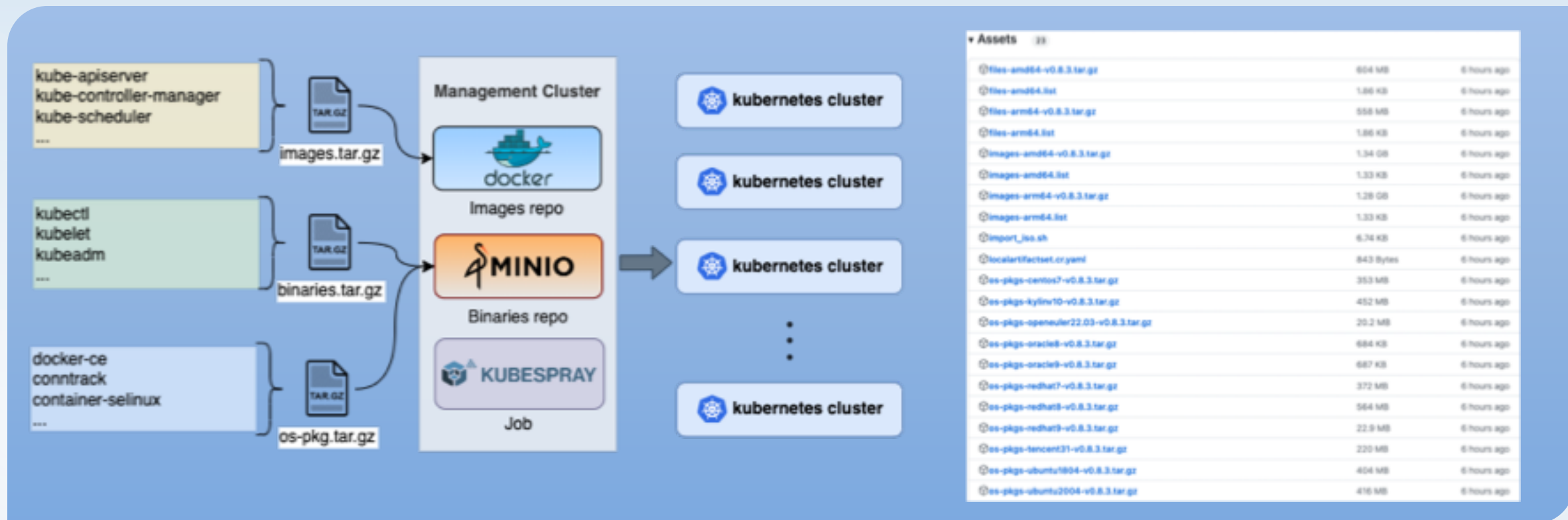
功能原理



发行版与架构支持情况:

- Linux 发行版：
 - 主流：Debian、Redhat、CentOS、Suse 等
 - 信创：统信UOS、银河麒麟V10、OpenEuler 等
- CPU 架构：
 - 主流：arm、传统 x86
 - 国产：海光、鲲鹏、飞腾

功能原理



便捷的离线化部署方案：

- kubean 发版时会通过 CI 生成一套默认版本的镜像、二进制、系统包等离线资源
- kubean 还提供了 airgap patch 方案，用户可按需生成指定版本的离线资源



Part 03



社区发展



社区发展



KUBESPRAY



yankay

67 commits 1,369 ++ 733 --

#16



ErikJiang

47 commits 1,119 ++ 770 --

#20



yanggangtony

24 commits 178 ++ 183 --

#30



tu1h

18 commits 180 ++ 127 --

#38



cyclinder

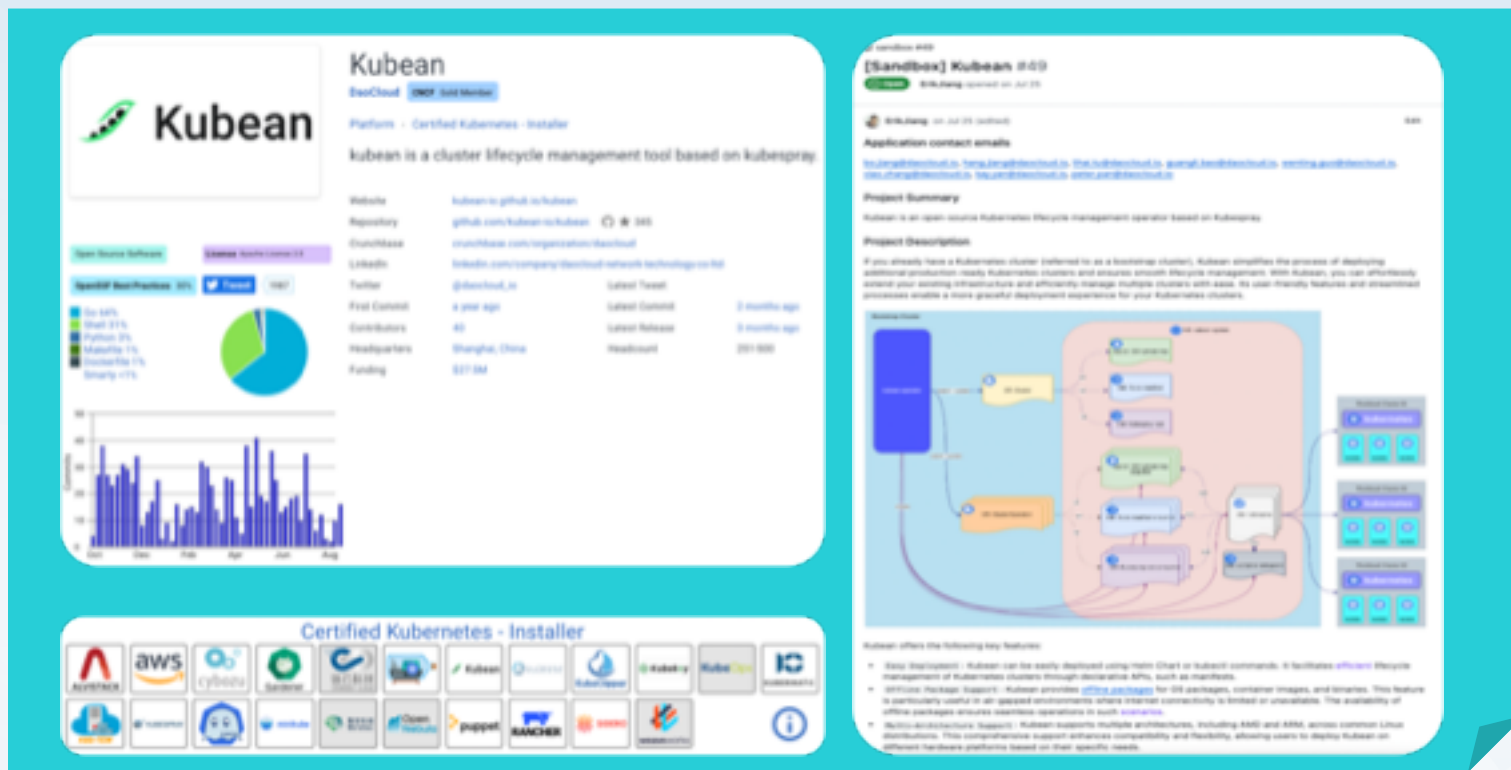
13 commits 138 ++ 115 --

#49

```
1 aliases:
2 kubespray--approvers:
3   - mattymo
4   - chadswen
5   - mirwan
6   - miouge1
7   - luckysb
8   - floryut
9   - eomichi
10  - cristicalin
11  - liupeng0518
12  - yankay
13  - mzaian
14 kubespray--reviewers:
15   - holsten
16   - bozzo
17   - eppo
18   - eomichi
19   - jayenlau
20   - cristicalin
21   - liupeng0518
22   - yankay
23   - cyclinder
24   - mzaian
25   - mrfreezeek
26   - erikjiang
27 kubespray--emeritus_approvers:
28   - riverzhang
29   - atoms
30   - ant31
31   - woopstar
```

- <https://github.com/kubernetes-sigs/kubespray/graphs/contributors>

社区发展



- <https://landscape.cncf.io/?selected=kubean>
- <https://github.com/cncf/sandbox/issues/49>



社区发展



Roadmap

- 支持更广泛的国产信创场景
- 持续优化 Kubean 的使用体验
- 允许扩展低版本集群的 LCM
- 社区建设及接纳更多元的开发者

社区发展



新手试玩



Kubean Playground



killercoda.com/kubean

Kubean Intro

Apply Kubean CR

Now we will create a cluster by Kubean, what a exciting moment!

We put the kubean-cr.yml into current directory in advance, you can check it out by `cat kubean-cr.yml`

Install the customised Kubean CR YAML, it will create a [Job](#):

```
kubectl apply -f ./kubean-cr.yml
```

Let's watch the logs of job that just launched:

The process takes a while, probably 15 minutes, you could grab some coffee~

```
sleep 5 && kubectl -n kubean-system logs -f $(kubectl -n kubean-system get pods -o jsonpath='{.items[0].metadata.name}') -l kubean-system
```

After all sucessfully done, we can simply check the cluster status:

```
export KUBECONFIG=/etc/kubernetes/admin.conf
kubectl get nodes
```

BACK

CHECK

Editor Tab 1

37 min

```
TASK [kubernetes/control-plane : Kubeadm | aggregate all SAs] *****
ok: [node1]
Friday 01 December 2023 07:41:54 +0000 (0:00:00.474) 0:14:01.610 *****
Friday 01 December 2023 07:41:54 +0000 (0:00:00.063) 0:14:01.673 *****
Friday 01 December 2023 07:41:55 +0000 (0:00:00.054) 0:14:01.727 *****
Friday 01 December 2023 07:41:55 +0000 (0:00:00.050) 0:14:01.778 *****
Friday 01 December 2023 07:41:55 +0000 (0:00:00.046) 0:14:01.824 *****

TASK [kubernetes/control-plane : Set kubeadm api version to v1beta3] *****
ok: [node1]
Friday 01 December 2023 07:41:55 +0000 (0:00:00.068) 0:14:01.892 *****

TASK [kubernetes/control-plane : Kubeadm | Create kubeadm config] *****
changed: [node1]
Friday 01 December 2023 07:41:56 +0000 (0:00:01.232) 0:14:03.125 *****
Friday 01 December 2023 07:41:56 +0000 (0:00:00.048) 0:14:03.173 *****
Friday 01 December 2023 07:41:56 +0000 (0:00:00.044) 0:14:03.217 *****
Friday 01 December 2023 07:41:56 +0000 (0:00:00.046) 0:14:03.264 *****
Friday 01 December 2023 07:41:56 +0000 (0:00:00.159) 0:14:03.424 *****
Friday 01 December 2023 07:41:56 +0000 (0:00:00.191) 0:14:03.615 *****
Friday 01 December 2023 07:41:56 +0000 (0:00:00.062) 0:14:03.678 *****
Friday 01 December 2023 07:41:56 +0000 (0:00:00.045) 0:14:03.723 *****
Friday 01 December 2023 07:41:57 +0000 (0:00:00.055) 0:14:03.778 *****
Friday 01 December 2023 07:41:57 +0000 (0:00:00.050) 0:14:03.829 *****

TASK [kubernetes/control-plane : Kubeadm | Initialize first master] *****
changed: [node1]
Friday 01 December 2023 07:42:10 +0000 (0:00:13.817) 0:14:17.646 *****

TASK [kubernetes/control-plane : Set kubeadm certificate key] *****
ok: [node1] => (item= --control-plane --certificate-key 4637603937ea752d8e6ab860c3634b38d3c90854ba33eeef173dfa819456395)
Friday 01 December 2023 07:42:12 +0000 (0:00:02.054) 0:14:19.701 *****
Friday 01 December 2023 07:42:13 +0000 (0:00:00.088) 0:14:19.790 *****

TASK [kubernetes/control-plane : Create kubeadm token for joining nodes with 24h expiration (default)] ***
ok: [node1]
Friday 01 December 2023 07:42:13 +0000 (0:00:00.258) 0:14:20.048 *****

TASK [kubernetes/control-plane : Set kubeadm_token] *****
ok: [node1]
Friday 01 December 2023 07:42:13 +0000 (0:00:00.070) 0:14:20.119 *****
Friday 01 December 2023 07:42:13 +0000 (0:00:00.047) 0:14:20.167 *****

TASK [kubernetes/control-plane : Kubeadm | Join other masters] *****
included: /kubespray/roles/kubernetes/control-plane/tasks/kubeadm-secondary.yml for node1
Friday 01 December 2023 07:42:13 +0000 (0:00:00.083) 0:14:20.250 *****
[]
```




诚挚欢迎更多的开发者
参与到 🌱 Kubean 社区的建设中

Github: <https://github.com/kubean-io/kubean>



Thanks.

起九生合
于層于抱
壘之毫之
土臺末木

